Gleacher Game Case Brief

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Background & Goals for the Future

The firm currently produces and sells a wearable heart monitor, which performs sophisticated, continuous diagnosis of the heart's health. The engineer who designed the product started the firm one year ago. He contributed his life savings of $4,000,000 to the firm and used the funds to purchase a factory for production and a distribution center to store and ship the monitors.

After one year of business, they are only selling about 10 monitors a day, so the engineer was about ready to pack it in when you and your team came along with an offer he could not refuse. Your team has paid him $2,000,000 for 50% of the firm and will take over the management of the firm. You will spend the next four years taking the business from a small shop to one positioned for a buy-out or an IPO. You will not be paid a salary, but hope to get a large payout down the road.

You will use your knowledge learned in life and at Booth to help you grow the firm. You must devise your growth strategy and determine the appropriate investments and operating changes you need to make to achieve your strategy. Operating levers at your disposal include finding new target markets, developing new products, optimizing your pricing structure, changing your operating technology and working with partners. You may choose to invest in new factories and distribution centers all over the world. You will finance these operating and investing activities through retained earnings, working capital management, long-term debt and an emergency line of credit.

Products & Development

Your firm's first product is a heart monitor that is integrated into an elastic band around one's chest. A miniature computer and transmitter are also integrated into the band, which allows the monitor's results and location to be quickly processed and uploaded over a 2.4 GHz wireless network to a server run your firm. The monitor records and monitors a detailed temporal profile of each heartbeat. As a result, custom software on the server can perform relatively sophisticated diagnoses including heart murmurs, heart valve problems, pending heart attacks, and over-exertion. Anyone with a password for the monitor can log into the server over the internet to view the results. The heart of your new technology is a signal processor that uses a neural network to detect even the faintest signals and the server software that analyzes the signal.

A variety of other monitors have become available that could be integrated with your core technology. The founding engineer's survey of the technology, coupled with insights from independent sales representatives resulted in a series of market research reports describing new technologies that the firm could use to create new products, as well as additional customer segments for the monitors.

In this simulation, a product design is defined by its set of features. To see the heart monitor features, click on the headquarters (HQ) icon and then the name of the product under the products heading. You will see the development time and development cost, as well as materials cost per unit that will be incurred when production begins.

You will be able to save a product design before starting development and incurring any development cost.

After a product design has been saved, you will be able to run a focus group to learn the willingness to pay (WTP) for the product from a random sample of potential customers in a given region and market. Each focus group consists of 10 participants and costs $20,000 ($2,000 per participant). The results of a focus group become available after 7 days.

After the development of a product design has been completed, you will see it appear in the product drop-down list for new shipping agreements on the factory panel.

Sales

Each distribution center (DC) includes sales representatives whose territories are limited to the distribution center's region. The sales representatives take orders 24 hours a day for products that are on hand in the distribution center. Once an order is received, the distribution center immediately ships the product. Revenue is recognized as soon as the unit ships. Payment will be received depending upon the customer and terms. Some customers may take longer to pay.

Sales representatives receive no base salary, but they are paid a commission in the amount of 20% of the sales price for retail sales. Commissions are recognized in the financial statements when sales are made.

There is also a handling cost of $10.00 per unit for retail sales from a DC.

Monitors are very reliable so returns are inconsequential and all customers eventually pay you, so there is no bad debt expense and no write-offs.

Each distribution center sets its own prices for its products. Prices can be changed at any time and the change takes place immediately.

Markets & Retail Customers

The world in the Gleacher Game consists of 7-10 regions, depending on the scenario.

Each region has one or more isolated markets. A market is a regional population of retail customers with similar needs or desires. Each customer's willingness to pay (WTP) is the sum of the values that she places on each of the product's features. The customer's consumer surplus is her WTP minus the retail price of the product. A given product from a given manufacturer sold at a given distribution is considered a separate stock keeping unit (SKU).

Retail customer arrivals follow the Bass model. In the Bass model, a small number of early adopters arrive to consider wearable diagnostic devices. If the early adopters purchase the monitor, word about the devices spreads and the arrival rate gradually increases as more consumers buy the monitor. Eventually as more and more of the customers in a market are satisfied, only late adopters are left, and the arrival rate of consumers decreases.

Pursuant to the Bass model, there are three types of arriving customers in the Gleacher Game: (1) innovators, (2) imitators, and (3) customers attracted through advertising.

(1) The number of arriving innovator customers is driven by a constant coefficient of innovation (p) multiplied by the remaining market size of unserved customers. When an innovator customer arrives, she checks all the products on hand in the distribution centers in her region (i.e., all SKUs) and determines her consumer surplus. If any of the SKUs provide a positive surplus for the customer, she purchases the item with the highest surplus and never returns.

(2) The number of arriving imitator customers is driven by a constant coefficient of imitation (q) multiplied by both the remaining market size of unserved customers and the proportion of the initial market size that has purchased the SKU for which they are imitators. When an imitator customer arrives, she only checks her consumer suplus for the SKU for which she is an imitator. If that SKU provides a positive surplus for the customer, she purchases it and never returns.

(3) The number of arriving customers due to advertising is driven by the amount of advertising spent in on a given day for a given SKU multiplied by both an advertising coefficient (p-adv) and the remaining market size of unserved customers. When a customer that has been attracted through advertising arrives, she only checks the SKU that was advertised to her. If that SKU provides a positive surplus for the customer, she purchases it and never returns.

Customers that cannot find a product with a positive consumer surplus return to the pool of unserved customers.

Please see the *Bass Model Estimation* section below for additional details. Information regarding each specific market, including market size, willingness to pay, and Bass Model coefficients, is provided in separate market research documents.

Bass Model Estimation

Q_t = arrivals at time *t*

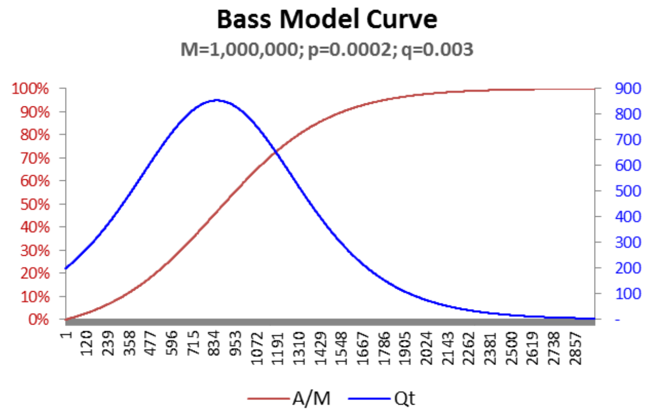
p = coefficient of innovation

q = coefficient of imitation

M = market size

A = cumulative number of sales to date

A/M = % of market served to date



To solve for the number of customers arriving at time *t*:

$$Q_t = [p + q*(A/M)]*(M-A)$$

If every customer that arrived were served, the time to peak sales would be:

$$t^* = [1/(p+q)] * [\ln(q/p)]$$

If every customer that arrived were served, then the size of peak sales would be:

$$s^* = M[(p+q)^2/4q]$$

As mentioned in the previous section, it is also possible to pay for advertising, which has the effect of an additional p-adv coefficient for a given SKU.

Gleacher Demographics

Geography

All customer segments are geographically isolated from each other. If a consumer buys a product in one region, it will not have any effect in any other region. Nine of the regions are similarly sized and their demographics are identical. Thus, if two of these regions started with the exact same distribution center with the exact same product at the exact same price, the two distribution centers would see the exact same demand until something changed for one of the distribution centers.

There may be one region experiencing a significant ongoing military conflict, in which case it would contain several military bases spanning all branches of the military. There would be a civilian population in this region as well, but possibly differing in magnitude from the population of the non-military regions.

Medicine

There are about a dozen small, independent segments consisting of different medical practices. A department or physician would purchase a monitor appropriate for their specific practice, provide it to a patient as part of that patient's treatment, and then review the collected data online. Selling a monitor to one segment would have no effect on the consumers in the other medical segments. The cancer segments (prostate, breast, kidney/bladder, lymph, and bone/blood) have the highest price points, some up to $2,000 per monitor, depending on the specific segment and product features. Other medical segments' price points are slightly lower, starting at hundreds of dollars but rarely above $1200.

Clinic

The clinical segments are similar to the medical segments except a clinician would provide the monitor to a patient but the patient, instead of a physician, would have primary responsibility for monitoring the data. There are two clinical segments: cardiovascular and fertility. Cardiovascular concerns are mainly blood pressure, heartbeat, and dissolved gasses in the bloodstream. A fertility monitor would be used to monitor reproductive hormones to provide insight into a woman's fertility over time. Price points are typically in the hundreds of dollars, depending on the features. The clinical segments tend to be larger than the medical segments.

Trainers

This segment consists of personal physical trainers selling monitors to their athlete clients. The primary interest would be a heartbeat monitor that operates continually. However, the trainers would also value other features related to general physical health. Unlike the other market segments, consumers in this market will pay a premium if the finish matches their personal taste. In fact, a small part of this segment would pay a substantial premium if there were a finish that was considered fashionable. A fashion segment could grow quickly but could also taper off quickly.

Law

In these segments, police departments and other government organizations purchase the monitors to measure alcohol or different families of drugs in the bloodstream of individuals, mostly parolees. This segment is composed of six independent segments. The five segments for different types of drugs tend to have slightly lower price points and similar sizes as the medical segment. The alcohol (ethanol) segment is about twice as large as a single drug segment, although the price point is at least a couple hundred dollars lower.

Military

As soldiers are deployed to places where chemical weapons are a concern, the military would like to outfit those soldiers with monitors, depending on which chemical or group of chemicals are suspected. There are four military segments: botulinum, anatoxin-a, sarin and cyclosarin, and soman. The segments are roughly similar in size with similar price points. These segments only reside in the possible military region. In terms of finish, there tends to be a slight preference for darker colors.

Creating Batches of Product

A batch is some number of units, all of the same product, that are shipped from one facility to another. If a batch originates at a factory, then the batch is created when the factory starts manufacturing it. If a batch originates in a distribution center, then the batch is created by taking units from available finished goods at the distribution center. All batches are shipped to distribution centers.

A firm can produce any of its developed product designs in any of its factories. Products are produced in batches, and once a batch is finished, it is shipped to a distribution center. A firm can ship to its own distribution center or to a distribution center owned by another firm. Once a batch arrives at a distribution center all the units are immediately available for sale. Production and distribution of a batch is triggered by a shipping agreement.

Production

Your firm's current factory is a large, mostly empty, building that employs four engineering contractors over multiple shifts 7 days a week. Production of a batch begins with the engineers procuring parts from a nearby supplier. The engineers then share general purpose equipment at a couple of lab benches to fabricate a batch of 100 monitors. Once the batch is completed, they are mailed to the firm's distribution center and the engineers start the next batch.

Every factory includes land that costs $100,000. The factory also includes some amount of plant and equipment chosen by the firm. The amount of plant and equipment is measured by the capital expenditure the firm pays for it. The factory becomes operational 90 days after construction begins. Additional plant and equipment can be purchased at any time. After 30 days, a capital expansion becomes operational as soon as the factory completes its current batch, or, if the factory is idle, immediately. Capital expenditures are irreversible and are straight-line depreciated over 15 years. Land does not depreciate.

Once a factory is operational, it begins incurring recurring costs including utilities, insurance, and wages. All these costs are aggregated into one overall daily recurring expenditure rate. The firm can increase or decrease a factory's daily expenditure at any time and the change takes effect as soon as the factory completes its current batch, if any. Setting a factory's daily expenditure to zero shuts the factory down and suspends all of its shipping agreements.

The time it takes to produce a batch depends on the factory's operational plant and equipment, its daily expenditure rate, the factory's production technology, and the number of units in the batch.

Yearly production is determined by the Cobb-Douglas function below:

$$Y = AK^{\alpha}L^{\beta}$$

Where:

- Y is the quantity produced over the year
- A is total factory productivity
- K is the units of capital employed
- L is the daily recurring costs * 364
- α is the coefficient on capital
- β is the coefficient on labor

Daily throughput (without setup time) is:

$$\lambda = 364^{(ß-1)} A K^{\alpha} l^{ß} = \frac{A K^{\alpha} (l * 364)^{\beta}}{364}$$

Where:

- λ is the daily throughput
- A is total factor productivity
- K is the units of capital employed
- l is daily recurring costs
- α is the coefficient on capital
- β is the coefficient on labor

Each technology has a different setup time for each new batch that must be incorporated into your cycle time and throughput calculations.

The minimum capital investment for a production line is $500,000 and the minimum capital investment for an automated cell is $3,000,000.

The table below summarizes the parameters for the different technologies.

Production Technologies

Technology	A	α	β	Technology Description
Benches	0.009	0.10	0.85	Lab benches with general-purpose equipment. Each table is manned by a skilled worker who produces an entire product. Setup time for new batches = 0.05 days
Production Line	0.010	0.30	0.75	A row of tables with specialized equipment and each worker assigned a specific portion of the production sequence. Workers are relatively unskilled. Setup time for new batches = 0.50 days
Automated Cell	0.020	0.80	0.30	A robot that feeds material to automated production machines clustered around it. There is support staff but no workers performing actual production. Setup time for new batches = 1 day

Product Costing

HQ financial statements are prepared using absorption costing. The factory's standard unit cost is the sum of the depreciation and recurring expenses that are incurred to produce one unit at the factory's current capital and recurring cost level, at a designated batch size. In year 1, the batch size was 100, but that can be changed, and should be changed, for certain technologies and strategies.

When a batch begins production, the cost of materials is immediately added to the work-in-process inventory account. Then at the end of each day the batch is in process, as well as at the moment the batch is completed, the incremental depreciation and daily expenditures incurred during production of the batch are added to work-in-process inventory.

Depreciation and other recurring expenses during factory idle times are designated as idle factory costs and are charged to income in the period incurred.

Distribution

Distribution centers hold finished inventory and employ a standard staff including a material handler, a logistics specialist, and a sales representative. There is no practical upper limit on the number of units that a distribution center can store.

It takes 60 days to build a new distribution center. As soon as a firm begins construction of a center, it spends $100,000 on land and $2,500,000 on the plant and equipment. Once the distribution center is operational it incurs a fixed recurring cost of $2,000 per day, not including fulfillment costs. DCs are straight-line depreciated over 15 years. Land does not depreciate.

You may shut down a distribution center if it has no inventory, either on hand or on the way to the distribution center. A distribution center that is shut down does not incur any recurring costs and all its shipping agreements are suspended. A shut down distribution center can be restarted at any time.

The distribution center interface includes different plots of sales and inventory. Inventory is classified as “on-hand,” which inventory physically in the center and available for sale, and “total”, which is on-hand inventory plus all inventory currently in transit to the distribution center, plus all work-in-process inventory that will ship to the center when completed. A shipping agreement is triggered when the total inventory is at or below the reorder point.

Shipping

There are two choices for how a completed batch is shipped to a distribution center: mail and container. It takes one day and costs $200 per shipment (10 units) by mail if the origin and destination are in the same region or three days and $400 per shipment (10 units) if the origin and destination are in different regions.

Shipping a container within the same region costs $5,000 and takes 7 days. Shipping a container between regions costs $10,000 and takes 21 days. A container can hold up to 1,000 units. So, if a batch consists of 1,000 units or less, the batch is shipped in one container. On the other hand, if a batch consists of 1,001 units, the batch is shipped in two containers, with both containers leaving the origin at the same moment.

If a shipping agreement results in multiple batches of the same product shipping from a distribution center to another at the same moment, then those batches are combined into one larger batch. For example, if a shipping agreement has a quantity of one but an order point that causes the shipping agreement to create 200 batches of one each unit to ship, those 200 batches of one unit each would be combined into one batch of 200 units. If the shipping mode in that case was a container, all 200 units would ship in the same container. On the other hand, batches created by different shipping agreements are shipped separately even if the batches are created at the same time.

Shipping agreements may only be initiated at the origin.

Shipping costs are paid by the origin.

Debt and Dividends

Trade Credit

All raw materials costs are paid at the end of the 30th day after the payable is incurred.

All other payables — recurring factory and distribution center costs, as well as shipping and fulfillment costs — are paid at the end of the 15th day after the payable is incurred.

Emergency Loan

If a payable comes due and there is insufficient cash to pay it, your company automatically borrows enough money from an unsecured emergency line of credit to the pay the debt at a 40% APR. Any cash that your company generates will automatically go toward paying down the emergency line of credit. As long as there is a positive balance on the emergency line of credit, your company will not be able to start development on new products, invest in capital expenditures, or engage in any other transactions that require cash to be spent up front.

Bonds

You may incur long-term debt by selling semi-annually compounded zero-coupon bonds in an open market. Each bond has $1,000 face value and a 5-year maximum term. The proceeds from the sale of the bond depend on how an independent rating agency rates your firm. Your firm might also be rated as not viable for new long-term debt, which would prevent your firm from selling bonds at all. Information on the panel includes your firm's current rating, yield to maturity, and number of bonds you can sell at that rating.

Each time you issue one or more bonds, the issue will show up in the *Bond Issue* table on the *Debt* panel. The table includes the *Redemption Price* of bonds in each issue. You may retire bonds at that price per bond. When you retire a bond, the redemption price times the number you retire is immediately subtracted from cash. All bonds will be retired at book value, regardless of whether interest rates have risen or fallen.

Your credit rating is determined by your interest coverage ratio. The interest coverage ratio is yearly EBIT/(imputed annual interest on your line of credit + imputed annual interest expense on bonds). Yearly EBIT is calculated as the EBIT for the last fully elapsed quarter * 4. EBIT is the same as Operating Income.

The simulation will automatically tell you how much debt you can borrow at "Excellent", "Good", and "Poor" credit ratings. The thresholds and interest rates are below. Interest is compounded semi-annually.

Rating	Threshold	Interest Rate (APR)
Excellent	20x	10%
Good	7x	15%
Poor	2x	25%

Dividends

You earn a 6.5% after-tax return on any dividends that are paid out to the shareholders.

Reports

The reports accessible through the Headquarters panel and the Open Reports dropdown to the top-right of the map are designed to help you analyze the performance of your company and its competitors.

Daily Plots

You may also plot a variety of daily data from the Headquarters, Factory, and Distribution Center panels, including various presentations of inventory, pricing, and sales. Once you have opened a plot, you can download the data, drag your mouse over a region of the plot to zoom in on that region, or click on an entry in the legend to just show that one line on the plot.

Financial Statements

The game contains an income statement for each factory and distribution center, as well as consolidated financial statements – balance sheet, income statement, and cash flow statement – for the entire firm. Each statement can be aggregated by day, week (7 days), quarter (91 days), or year (364 days).

Most cash transactions, including converting receivables and payables, and receiving and paying interest, are processed only at the end of each day. There are other transactions, however, that affect cash immediately, including issuing and retiring debt, and starting construction or expansion of facilities.

Revenue and expense accounts are closed out to retained earnings at the end of each day as well.

Scoreboard

The game scoreboard will reflect your ranking as measured by Return to Investors, Retained Earnings, and Cash.

Return to Investors

= cash on hand - debt + dividends paid + after tax return on dividends distributed

Retained Earnings

= sum of all net earnings less dividends declared

Cash Position

= cash on hand - emergency loan